UC Love Data Week

February 13, 2026

Critical Integration of GenAI in Qualitative Analysis

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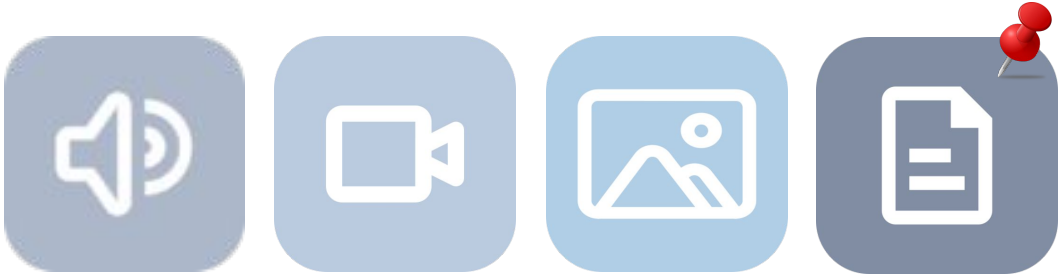
UCSB Library - Research Data Services

Recordings and assets will be soon available at: https://bit.ly/UC-LDW

Learning Objectives

- **Situate** current debates around the use of GenAI in qualitative research
- **Introduce** the Guided AI Thematic Analysis (GAITA) framework for effective and responsible use
- **Demonstrate** an integrated GenAI-supported analysis workflow

Qualitative Data



interviews
focus groups
case studies
observations/ethnography
open-ended surveys
compilations (literature/documents)
web scraping

Qualitative Data Analysis - Coding

Qualitative coding involves **labeling, categorizing, and organizing** qualitative data to **uncover patterns, relationships, and themes**.

By breaking raw text **into meaningful, manageable segments**, it serves as a foundational step in qualitative analysis and supports the **rigor and credibility** of the research.

It is not merely a mechanical task; rather, it is **an iterative, reflexive, and interpretive practice refined through theoretical insights and engagement with the dataset itself.**

Approaches

Inductive Coding
(The "Bottom-Up" Path)

A combined approach (blended path) can be used to balance structure and data-driven discovery

Deductive Coding
(The "Top-Down" Path)

EXISTING FRAMEWORKS
Established Theories, Frameworks, Protocols

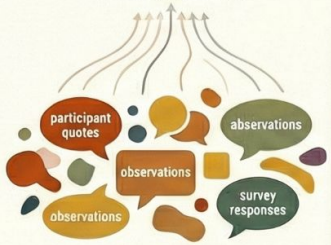
Ideal for Unexplored Territory
Best for identifying unexpected insights and exploring understudied topics or populations.

Data-Driven Discovery
Codes are created based on patterns found directly within the raw data.

From Observation to Theory
Start for findings inow the cruvnets.

Researchers start with raw data, group findings into categories, and build themes.

Discovery/Exploration



'RAW DATA'
Raw Data, Field Notes, Transcripts

Theory-Driven Analysis
Analysis begins with a pre-determined codebook based on existing theories or frameworks.

Roadmap/Structure

Systematic & Focused Testing
Best for testing, expanding, or comporing data against established research and protocols.

From Theory to Data

Researchers start with pre-defined codes and locate specific data excerpts that fit.

Comparison of coding styles using a workplace stress example

Feature	Inductive (Example)	Deductive (Example)
Example Code	"Missing Transition Time"	"Work-Home Boundary issues"
Logic	Derived from participant's exact phrasing.	Aligna with established work life balance theories.
Outcome	Captures unique, verbatim nuances.	Connects findings to broader academic frameworks.

Infographic generate by Google Notebook based on: ucsb-library-research-data-services.github.io/qualdata-training.

Qualitative Data Analysis Software (QDAS)

NVIVO

MAXQDA

TAGUETTE

ATLAS.ti

dedoose

QualCoder

Powered QDAS Tools

Platform	Type	Model/Provider Options	Data Privacy & Use	Researcher Control
NVivo & Atlas.ti	Proprietary	OpenAI (enterprise agreement), model versions not disclosed	Zero data retention; AI tasks do not train models	Vendor-managed; AI applied on-demand; no model/prompt customization
MaxQDA	Proprietary	Third-party LLMs (Claude, Gemini), cloud-based	Zero data retention; encrypted transmission	Vendor-managed; user controls tasks, not models
QualCoder	**Open Source**	**Several options, local and remote LLMs**	**Can run fully offline; remote AI optional**	**Full user control: choose your model, configure prompts, control AI usage**

An Open Debate

- **Ubiquitous:** embedded across most applications
- **Disruptive:** demands new ethics, rules, and methods
- **Questionable reliability:** prone to bias and hallucinations
- **Plasticity:** rapid adaptation

Researchers are using it with little support or guidance.

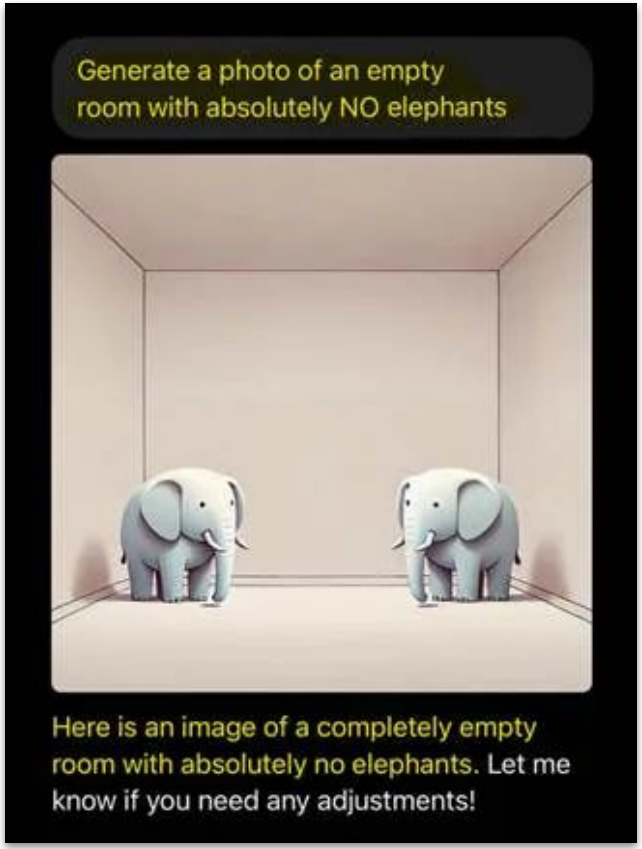


Image from: Brahmbhatt, A. (2025, March 3). *No elephants in the room? AI seems to think otherwise*. Medium. https://medium.com/@avanib28264/no-elephants-in-the-room-ai-seems-to-think-otherwise-7d70d6a7d5a4

The Opponent View

Qualitative Inquiry

Impact Factor: **1.8** / 5-Year Impact Factor: **2.2**

Open access | Research article | First published online December 17, 2025

We Reject the Use of Generative Artificial Intelligence for Reflexive Qualitative Research

Tanisha Jowsey, Virginia Braun, [...], and Michelle Fine (+2) View all authors and affiliations

OnlineFirst | https://doi.org/10.1177/10778004251401851

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Abstract

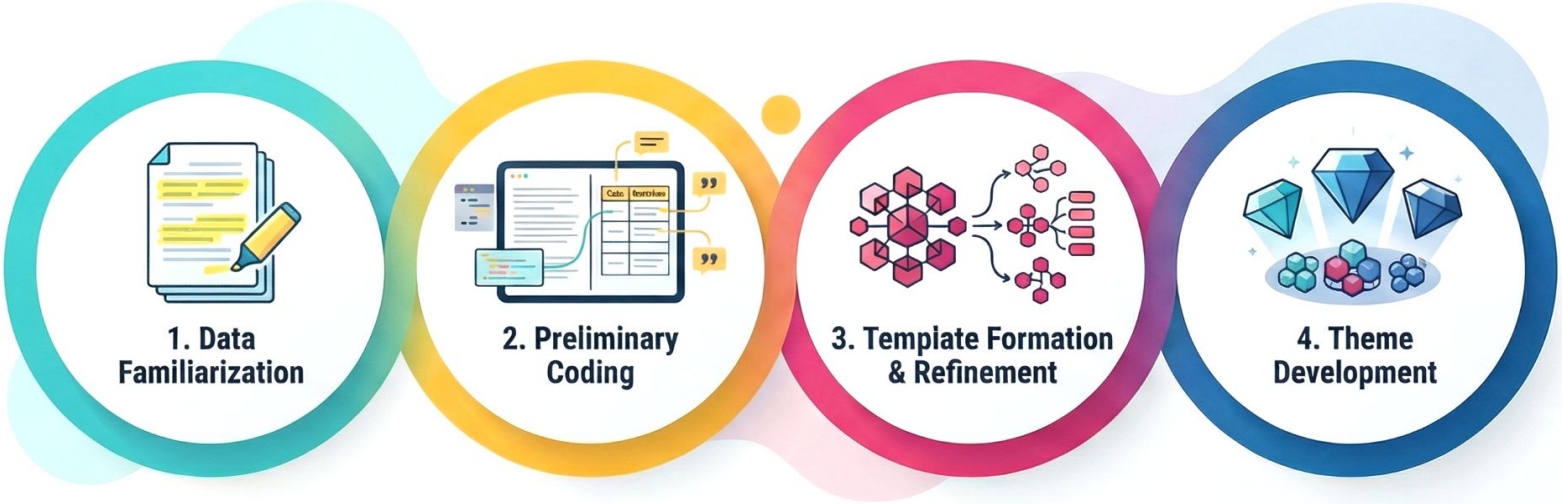
Four hundred and nineteen experienced qualitative researchers from 32 countries invite readers of *Qualitative Inquiry* to consider their position on use of generative artificial intelligence (GenAI) for qualitative research. We hold the position that analytic approaches such as reflexive thematic analysis are human research practices requiring a subjective, positioned, and reflexive researcher and therefore the use of GenAI in such approaches is not methodologically congruent. We additionally reject GenAI for reflexive qualitative approaches on the grounds of social and environmental justice.

Graphic generated by Google NotebookLM based on: Jowsey, T., Braun, V., Clarke, V., Lupton, D., & Fine, M. (2025). We Reject the Use of Generative Artificial Intelligence for Reflexive Qualitative Research. *Qualitative Inquiry*, *0*(0). https://doi.org/10.1177/10778004251401851

The Cautious Explorer

- Limitations and inconsistencies (e.g., context window limits, bias, hallucinations)
- Existing frameworks for ethical and responsible adoption (i.e., disclosure, traceability, and audit trails)
- Keeping humans-in-the-loop (i.e., closely supervised, smart querying, no automatic coding)

GAITA Framework: Workflow



Explore and summarize the dataset.

Elicit possible codes and subcodes from the data.

Group codes into clusters.

Identify higher-level, overarching **themes**.

Foundational Toolsets

PERFECT (Iterative Cycle) | **ACTOR** (Prompting Strategy) | **LEAP-PVD** (Ethical Framework)

Diagram partially generated by Google NotebookLM based on: Nguyen-Trung, K. (2025). ChatGPT in thematic analysis: Can AI become a research assistant in qualitative research? *Quality & Quantity, 59*, 4945–4978. https://doi.org/10.1007/s11135-025-02165-z

GAITA Framework: Iterative Cycle

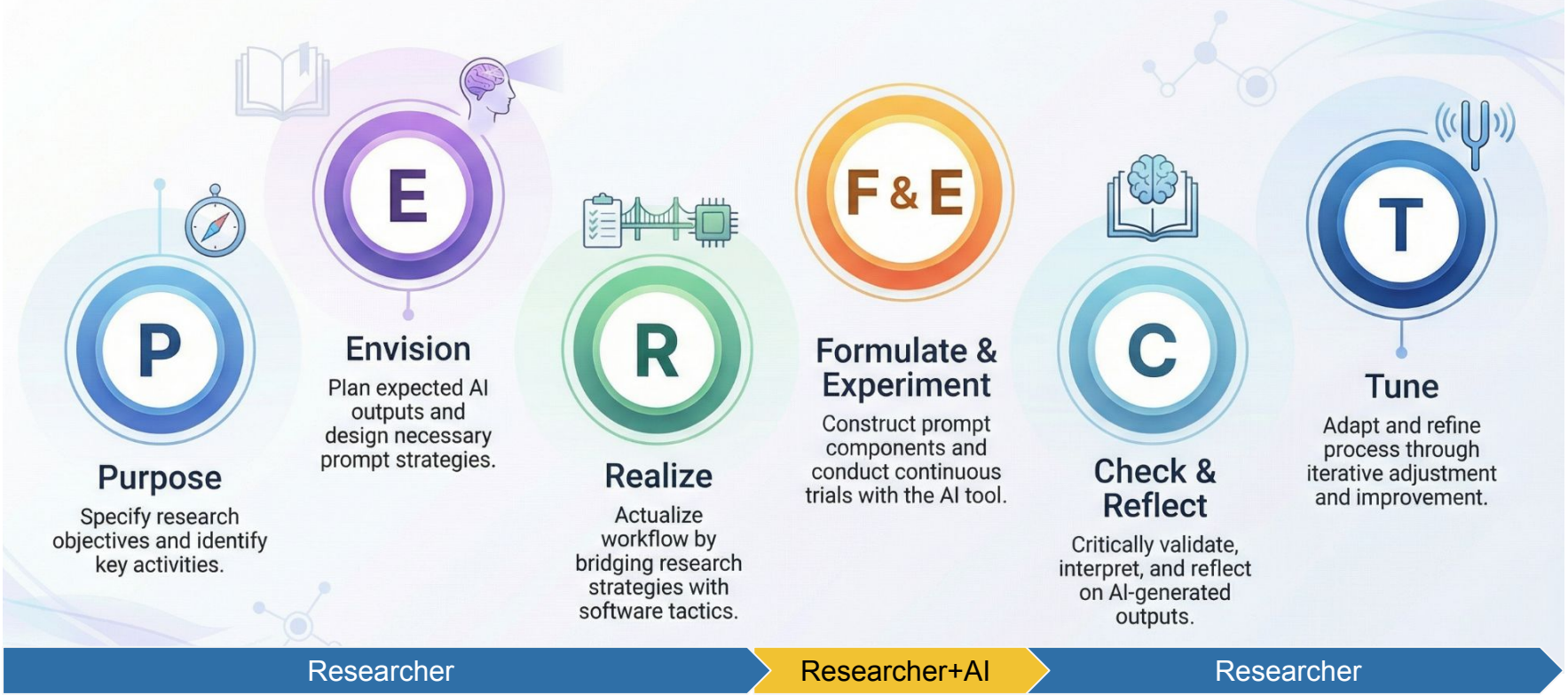


Diagram partially generated by Google NotebookLM based on: Nguyen, K. (2025). Generative AI in Qualitative Data Analysis: Introducing the Guided AI Thematic Analysis (GAITA) framework. The 2025 IASSIST Workshop Series: Tools for Qualitative Research, Online. Zenodo. https://doi.org/10.5281/zenodo.17872630

Any problems with this prompt?

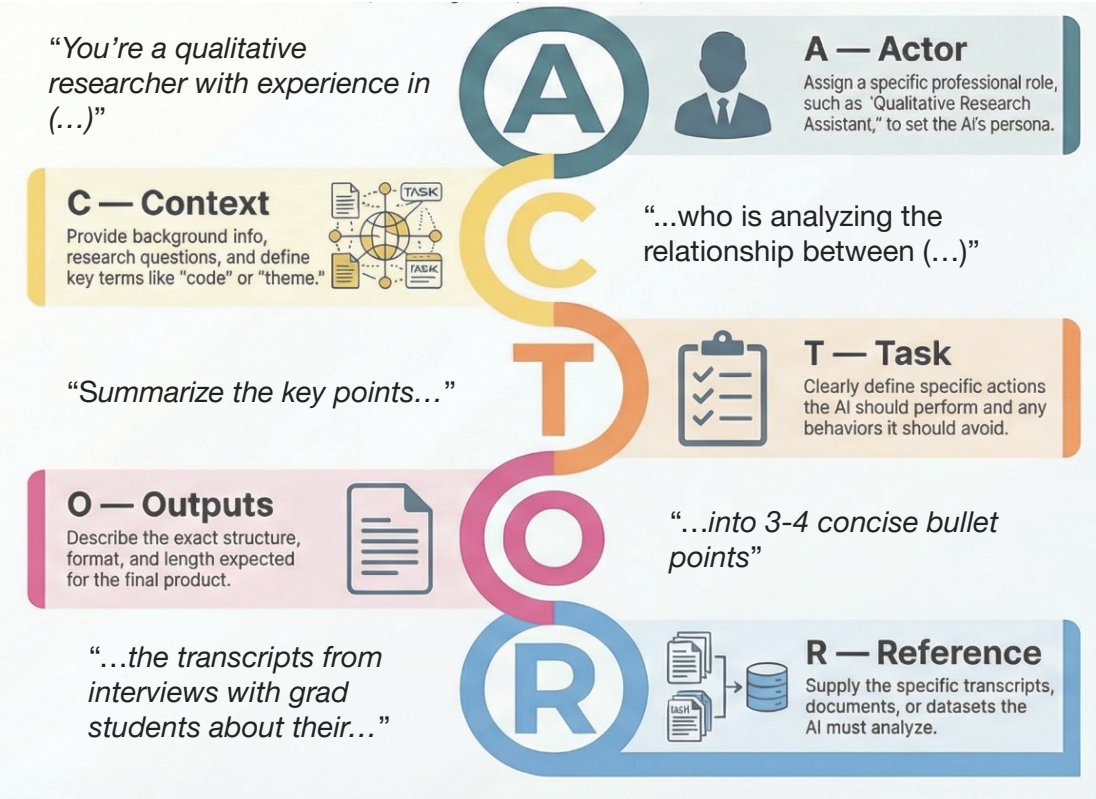
Undertake thematic analysis of the dataset I have uploaded alongside this prompt. Review the dataset in its provided format. Begin by ***familiarizing yourself*** *with the content and noting key recurring ideas.* ***Generate initial codes*** *based on identified features and apply these codes consistently across the dataset.* ***Group related codes into potential themes, using pattern detection*** *algorithms if applicable.* ***Evaluate and refine the themes*** *to ensure they are coherent and relevant. Develop detailed descriptions and clear names for each theme.* ***Prepare a comprehensive report summarizing the thematic findings****, supported by relevant data excerpts. Ensure consistency in coding and theme development,* ***validate the accuracy*** *of themes, and document the entire process transparently.* ***Provide between four and eight themes****, supporting the themes with the number of participants that are represented in each theme.* ***Provide quotes from participants*** *to support each theme.* ***Provide this thematic analysis summary*** *within 800 words.*

FAILED

Prompt from:
Jowsey, T., Stapleton, P., Campbell, S., Davidson, A., McGillivray, C., Maugeri, I., Lee, M., & Keogh, J. (2025). Frankenstein, thematic analysis and generative artificial intelligence: Quality appraisal methods and considerations for qualitative research. PLOS ONE, 20(9), e0330217. https://doi.org/10.1371/journal.pone.0330217

GAITA Framework: Prompting Strategy

- Structure prompts incrementally
- LLMs as actors, not thinkers (role-based prompts)
- Balance granularity
- Triangulation



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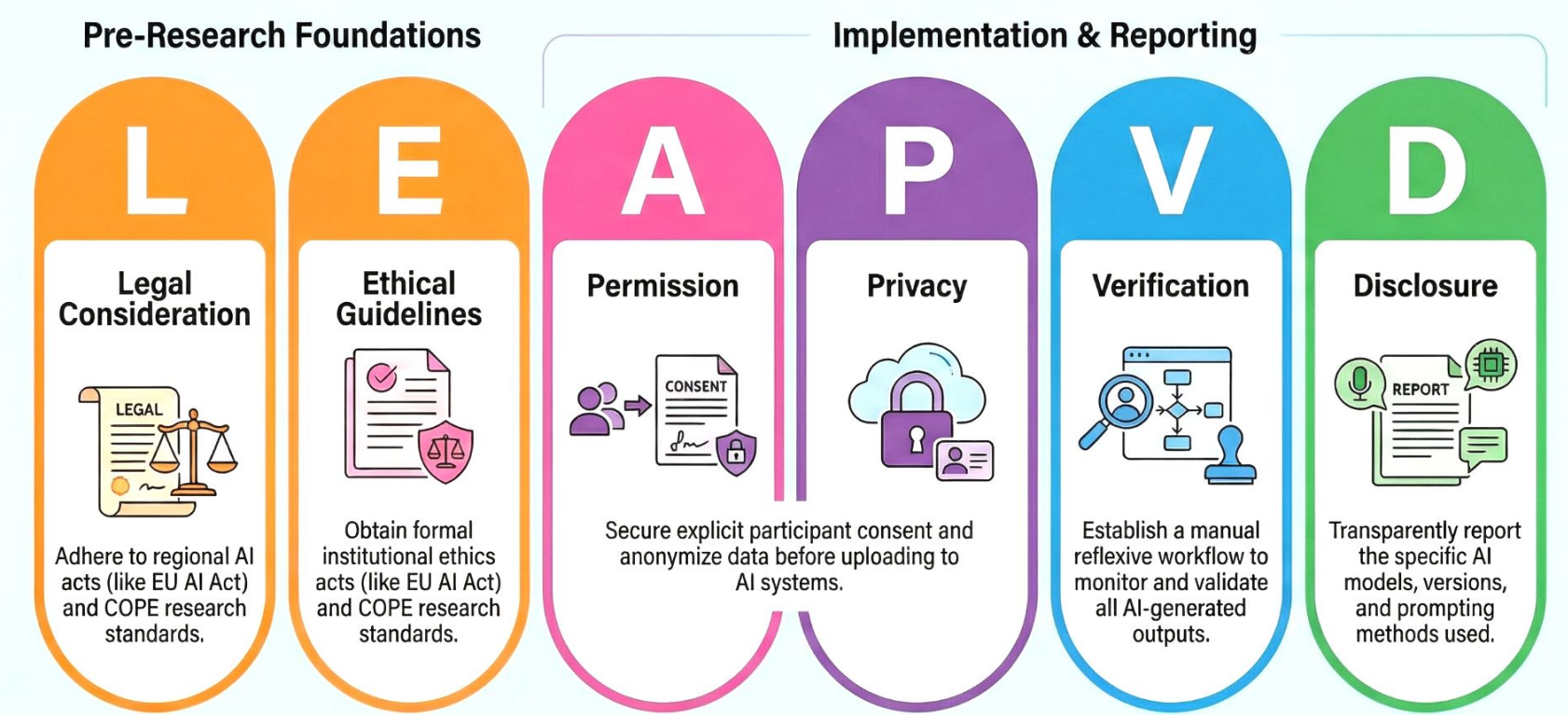
GAITA Framework: Ethical Considerations



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In Summary

- All Maintain control: Always adopt a human-first, human-centered approach
- All Iterative prompt refinement: Adjust prompts incrementally using best practices
- All Robust references: Include memos and definitions to support outputs
- All Thorough documentation: Record prompts, outputs, and key determinations
- All Workspace consistency: Avoid moving work across different workspaces
- All Transparency: Use declaration, specify the model used and the intended tasks
- Data privacy: Always de-identify* the data before analysis
- Data retention: Prioritize models that run locally
- Compliance and consent: ensure tools are IRB-compliant and prior consent is obtained.

Human Subject Research

*Curty, R. G. (2025, February 12). Safeguarding qualitative data with QualiAnon. UC Love Data Week, Online. Zenodo. https://doi.org/10.5281/zenodo.14861920

A Promising Match (Demo)

QualCoder

Open Source & Free
Multiplatform
Comparable to proprietary solutions
Growing user community

Ollama (llama 3.1)
Open Source & Free
Runs locally (safer setup)

https://doi.org/10.5281/zenodo.18604273